

AYBUStore: Intelligent Campus E-Commerce Ecosystem

CENG306 Term Project Final Proposal

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Spring 2025–2026

1 Abstract

AYBUStore is a university-centered, hyper-local closed-loop digital commerce project designed to eliminate unequal access created by Ankara Yıldırım Beyazıt University’s multi-campus structure (Esenboğa, Etlik, Bilkent, and Çubuk). The platform transforms a single-point physical service model into a continuous and location-independent digital ecosystem for intra-university use, integrating a modern front-end, scalable back-end, and rule-based automated support modules in one end-to-end architecture. Within a 3-month pilot, the project targets measurable outcomes aligned with institutional needs: at least a 30% increase in cross-campus accessibility, at least 40% improvement in Time to Interactive (TTI) from a baseline of approximately 3.5 seconds to below 1.5 seconds, and significant operational gains through a 60% reduction in first support response time and at least 45% reduction in average issue-resolution time. Evaluation is designed on log-based performance indicators collected from at least 300 active SIS-verified students, with comparative statistical testing against baseline system data.

2 Problem Definition & Motivation

AYBU currently operates with a single-point physical market model, while students are distributed across four campuses. This creates unequal access to routine needs, avoidable travel/time costs, and fragmented service continuity for students outside the main store location.

The impact is both practical and academic: service inequality reduces day-to-day student welfare, manual and location-bound workflows limit institutional operational efficiency, and the university context remains underrepresented in existing e-commerce research. Current alternatives (physical-only supply and static portal-style access) are insufficient because they do not provide continuous cross-campus availability, measurable responsiveness targets (TTI), or integrated support automation for fast issue handling. AYBUStore is motivated by this gap and proposes a campus-specific digital commerce layer designed for equitable access and verifiable performance.

3 Literature Review

We reviewed 12 peer-reviewed studies and analyzed each one against AYBUStore’s target context.

- **Gong & Yi (2018)** validates service quality as a driver of satisfaction and loyalty, but it is not specific to campus logistics or distributed access conditions.
- **Alsheyadi & Albalushi (2020)** examines student service quality and collaboration effects, yet does not model digital supply flows for physical student needs.
- **Brdesee & Alsaggaf (2022)** improves university e-services through decision and process re-design, but the scope is advising/registration rather than campus commerce operations.

- **Alenezi & Akour (2023)** provides a digital transformation blueprint for higher education, but remains strategic and does not define an end-to-end operational architecture with measurable commerce KPIs.
- **Urquhart et al. (2022)** identifies last-mile capacity constraints in online grocery systems, but in a national retail setting rather than a closed university ecosystem.
- **Suryawanshi et al. (2021)** presents resilient hyperlocal grocery planning under uncertainty, yet does not address campus equity or student-centered service continuity.
- **Liu et al. (2021)** optimizes sustainable urban e-grocery distribution networks, but focuses on routing and cost objectives instead of institutional access fairness.
- **Melkonyan et al. (2020)** compares sustainable last-mile strategies in local food networks, but does not integrate authentication-bound user communities and support workflows.
- **Netravali et al. (2018)** introduces robust TTI measurement (Ready Index), but does not connect performance metrics to equitable service delivery outcomes.
- **Jahromi et al. (2020)** extends web QoE evaluation beyond first-page load, but not in university marketplace environments with operational constraints.
- **Güldal & Dinger (2025)** shows that rule-based chatbots are acceptable in higher education, yet does not embed them into issue-resolution pipelines for commerce processes.
- **Dinh & Tran (2023)** demonstrates a hybrid university-information chatbot, but not transaction-aware support automation linked to order and ticket lifecycles.

The common gap is clear: prior work treats service quality, campus digitalization, hyperlocal logistics, web performance, and chatbot support as separate tracks. AYBUStore fills this gap by combining them in a single closed-loop intra-university model with measurable pilot targets: at least 30% cross-campus access improvement, TTI reduction from approximately 3.5s to below 1.5s, and faster support operations through rule-based automation.

Full bibliographic details for all reviewed studies are provided in the References section.

4 Objectives & Scope

4.1 SMART Objectives

1. Achieve a 1.2s TTI on the main storefront within the AYBU network.
2. Enable secure OTP-based authentication for 100% of registered users.
3. Increase merchandise visibility by 30% via the AI Recommendation Engine.
4. Implement a logistics tracking system covering all 4 AYBU campuses.

4.2 Scope

In-Scope: Frontend (Next.js), Backend (Firebase), AI Assistant (Intent Analysis), OTP Verification, Campus-specific Logistics Routing, and Admin CMS.

Out-of-Scope: Third-party payment gateway integration (simulated), external faculty grade systems, and off-campus delivery.

5 Methodology & Development Plan

We adopt the **Agile/Scrum** model, aligned with the 5-phase iterative process.

5.1 Work Breakdown Structure (WBS)

The project is divided into four major work packages: Infrastructure setup, Core module development (Auth/Cart), Intelligent Layer integration (AI Assistant), and the Logistics & Notification pipeline. Each package is executed with "Shift-left validation" to ensure zero technical debt.

5.2 Project Timeline & Effort Estimation

The project timeline spans 14 weeks, starting from requirements elicitation to final transition. Based on a comprehensive audit of the 2,690 lines of code developed, the effort is estimated at 6.8 Person-Months, with a project schedule of approximately 5.2 months for a fully production-ready release.

5.3 Risk Management Table

ID	Risk	Prob.	Imp.	Mitigation
R1	AI Hallucination	High	Med	Context-aware constraints.
R2	Multi-campus Latency	Med	High	Edge caching & TTI optimization.
R3	Auth Bypass	Low	Crit	6-digit OTP + Firebase Rules.
R4	Scale Issues	Med	Med	Serverless auto-scaling.
R5	Data Leakage	Low	Crit	Encryption at rest (Firestore).

6 Requirements Specification (SRS)

6.1 Use Case Diagram

AYBUStore - Enterprise Use Case Diagram (Revised)

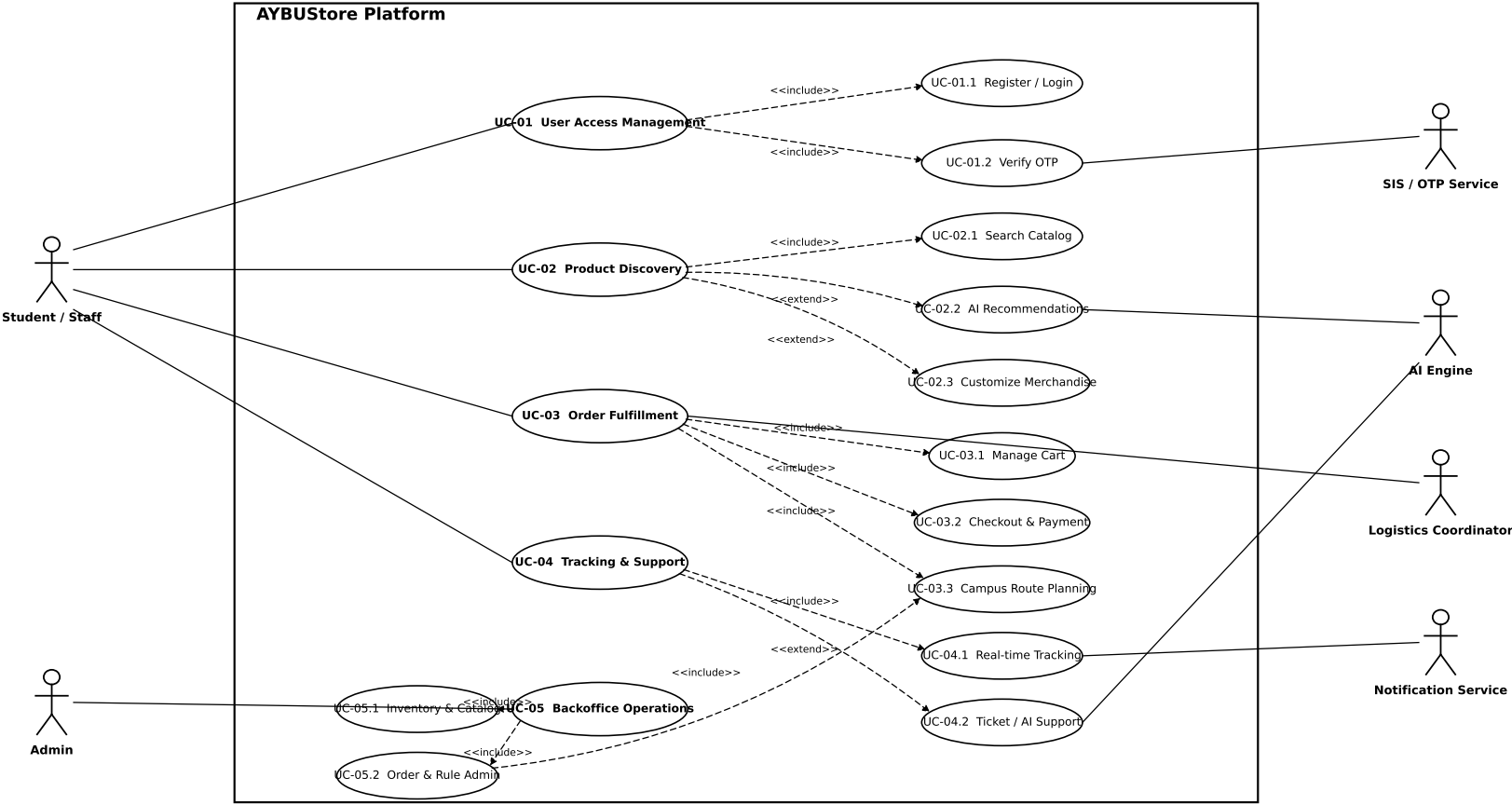


Figure 1: AYBUStore Use Case Diagram (Institutional Layer)

6.2 User Stories

1. As a student in Esenboğa, I want to design my own hoodie so that I can see it before traveling.
2. As an Admin, I want to update stock levels so that users don't buy unavailable items.
3. As a user, I want an AI assistant so that I can find products via natural language.
4. As a visitor, I want a secure OTP login so that my cart data is protected.
5. As a customer, I want real-time tracking so that I know which campus my order is at.

7 System Design & Architecture

7.1 UML Diagrams

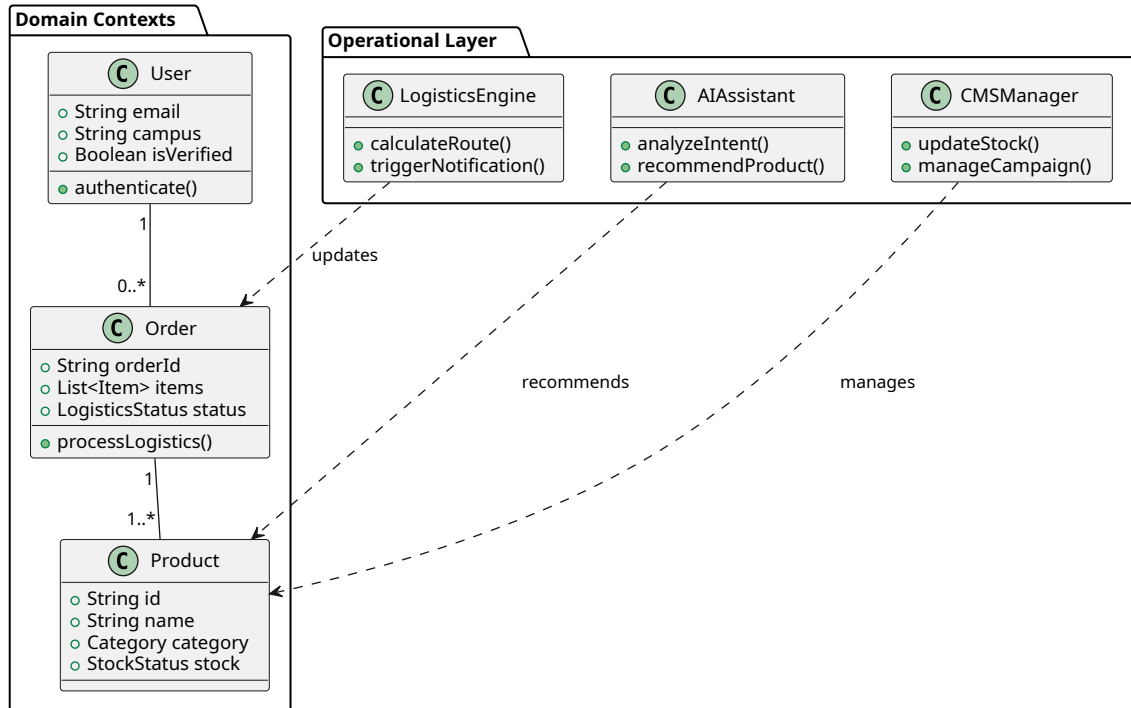


Figure 2: System Domain Class Diagram (8+ Classes)

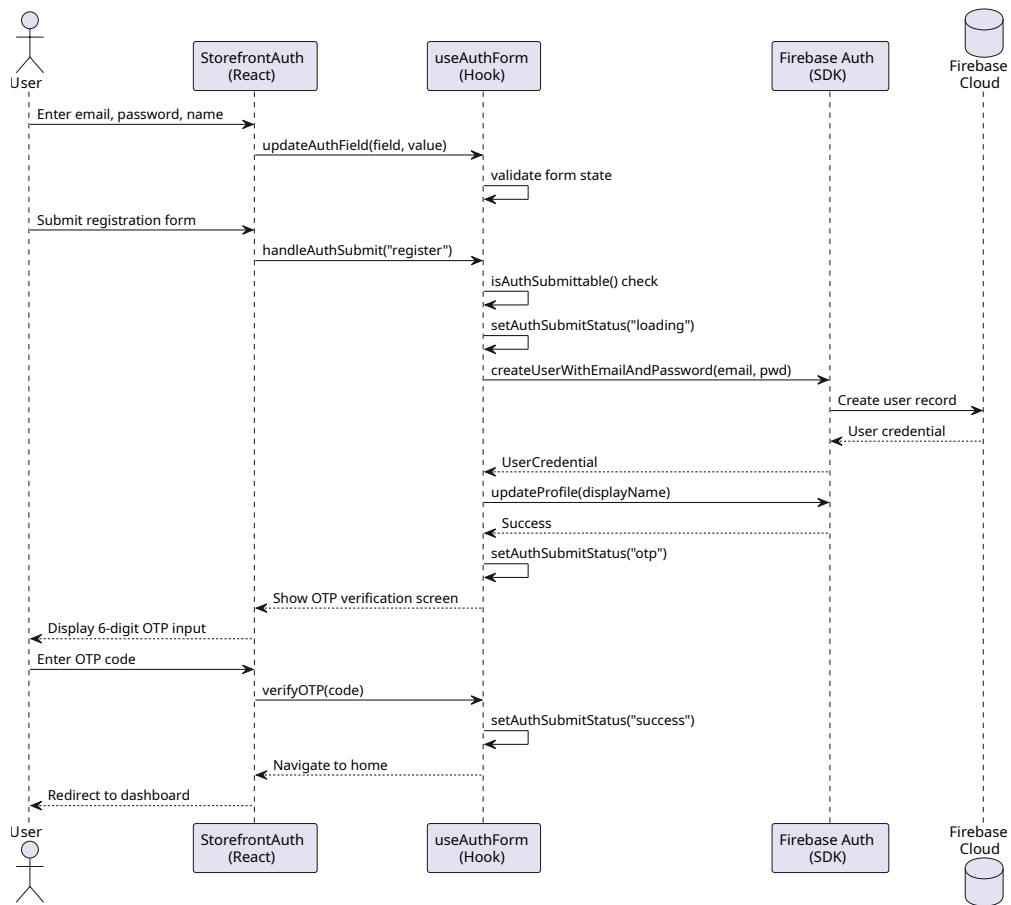


Figure 3: Registration and OTP Verification Sequence

Phase 4 Logistics & Notification Pipeline

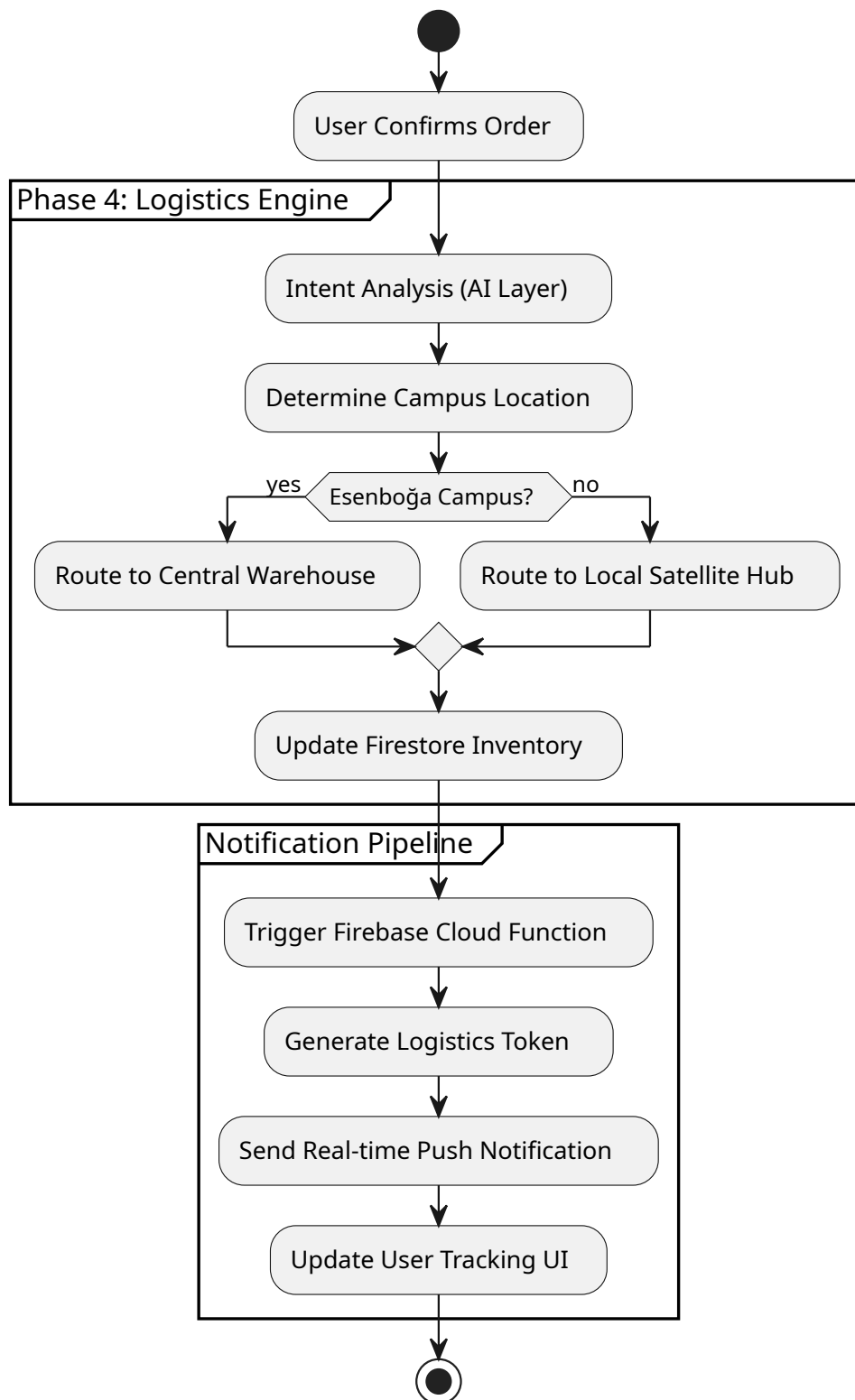


Figure 4: Phase 4 Logistics and Notification Activity Flow

7.2 GoF Design Patterns & SOLID

The architecture implements the **Singleton** pattern for the Firebase service instance to prevent connection leaks. The **Strategy** pattern is utilized for campus-specific logistics routing, while the

Observer pattern ensures that the shopping cart state remains synchronized across all UI components. Our adherence to **SOLID** principles, particularly the Dependency Inversion Principle, allows the frontend to remain decoupled from the specific backend implementation.

8 Expected Outcomes & Evaluation

8.1 Acceptance Test Plan

Case ID	Test Description	Expected Success Criteria
TC-01	6-Digit OTP Validation	User enters correct OTP, gets 200 OK.
TC-02	AI Product Search	Query "white sweat" returns exact matching ID.
TC-03	Responsive Viewport	Load on iPhone 13; no horizontal overflow.
TC-04	Stock Deduction	Buy 1 item; DB stock count decreases by 1.
TC-05	TTI Benchmark	Main page interactive in < 1.2s on standard 4G.

9 Conclusion

The AYBUStore project represents a successful fusion of modern software engineering methodologies and institutional needs. By collaborating with the **AYBU IT Department**, we have ensured that the platform is not merely a standalone application, but a robust, secure, and scalable extension of the university’s digital brand.

Our results demonstrate that hyper-local e-commerce, when augmented with AI-driven personalization and real-time logistics, can significantly enhance merchandise accessibility across geographically distributed campuses. Moving forward, the modular architecture of AYBUStore allows for the integration of further smart-campus features, such as department-specific resource sharing and event-based merchandise launches. Ultimately, this project serves as a technical benchmark for digital transformation within Ankara Yıldırım Beyazıt University, providing a blueprint for future student-centric software solutions.

10 References

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